Retrieve-then-Generate: Augmenting Language Models with Retrieved Passages for Knowledge-Intensive Tasks

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Abstract

RetGen combines retrieval and generation for Open-domain QA. It first uses DenseRet to retrieve passages, then conditions a Seq2Seq Generator on the question and the retrieved evidence. This division lets the generator consult current evidence instead of relying only on memorized parameters. We evaluate the pipeline on Natural Questions and HotpotQA, separating single-step evidence selection from the harder multi-hop setting. RetGen improves Exact Match over a closed-book generator on both datasets, with a larger gain on Natural Questions and a smaller gain on HotpotQA. The results show that retrieval can make answers more grounded while also exposing a remaining challenge: a generator must connect multiple retrieved facts when the answer depends on more than one passage.

Keywords

Retrieval-augmented generation, Dense retrieval, Natural Questions, HotpotQA

1. Introduction

Open-domain QA requires both finding evidence and expressing an answer. A generator without retrieval may produce fluent but unsupported responses, while a retriever alone does not synthesize an answer. RetGen addresses this interface by pairing DenseRet with a Seq2Seq Generator. We evaluate Natural Questions and HotpotQA to test both direct and multi-hop evidence use. DenseRet supplies the retrieval component, and the present paper extends that method as [1].

2. Method

For each question, DenseRet ranks passages by vector similarity and returns the top k candidates. The question, a compact separator, and the selected passages are concatenated into the input of a Seq2Seq Generator. Cross-attention allows the generator to copy or combine evidence while a source marker preserves passage boundaries. During training, the target answer is paired with a retrieved context; hard negatives are included by replacing one relevant passage with a high-scoring distractor. At inference, the retriever is frozen and the generator produces a short answer with evidence-aware decoding. The pipeline exposes retrieval depth as a controllable cost parameter. We use the same configuration on Natural Questions and HotpotQA so changes in Exact Match reflect the value of conditioning on retrieved passages rather than task-specific tuning.

3. Experiments

RetGen obtains 47.8 EM on Natural Questions compared with 35.7 for a closed-book generator, a gain of 12.1 points. On HotpotQA it reaches 31.2 EM versus 27.5, reflecting the difficulty of joining multiple facts. Increasing retrieval depth helps direct questions but gives diminishing returns on multi-hop questions when relevant passages are not jointly selected.

System	Natural Questions EM	HotpotQA EM
Closed-book generator	35.7	27.5
RetGen	47.8	31.2

4. Conclusion

RetGen shows that DenseRet retrieval and a Seq2Seq Generator form a useful evidence-to-answer interface for Open-domain QA. The 12-point Natural Questions improvement supports retrieval grounding, while the smaller HotpotQA gain identifies multi-hop composition as the main limitation. Better evidence planning is the next step.

References

[1] Vik, A. (2020). DenseRet: Learning Dual-Encoder Representations for Open-Domain Passage Retrieval. Conference on Natural Language Systems (CNLS).